

DIRECTIONS for questions 1 to 4: The passage given below is accompanied by a set of **four** questions. Choose the best answer to each question.

For Hegel, the contradictions that drive history exist first of all in the realm of human consciousness, i.e., on the level of ideas – in the sense of large unifying world views that might best be understood under the rubric of ideology. Ideology in this sense is not restricted to the secular and explicit political doctrines we usually associate with the term, but can include religion, culture, and the complex of moral values underlying any society as well.

Hegel did not believe that the real world conformed or could be made to conform to ideological preconceptions of philosophy professors in any simpleminded way, or that the "material" world could not impinge on the ideal.

For Hegel, all human behaviour in the material world, and hence all human history, is rooted in a prior state of consciousness. This consciousness may not be explicit and self-aware, as are modern political doctrines, but may rather take the form of religion or simple cultural or moral habits. And yet this realm of consciousness in the long run necessarily becomes manifest in the material world, indeed creates the material world in its own image. Consciousness is cause and not effect...

Hegel's idealism has fared poorly at the hands of later thinkers. Marx reversed the priority of the real and the ideal completely, relegating the entire realm of consciousness – religion, art, culture, philosophy itself – to a "superstructure" that was determined entirely by the prevailing material mode of production... Paul Kennedy ascribed the decline of great powers to simple economic overextension... But whether a highly productive modern industrial society chooses to spend 3 or 7 percent of its GNP on defence rather than consumption is entirely a matter of that society's political priorities, which are in turn determined in the realm of consciousness.

The materialist bias of modern thought is characteristic not only of people on the Left who may be sympathetic to Marxism, but of many passionate anti-Marxists as well. Indeed, there is on the Right what one might label the Wall Street Journal school of deterministic materialism that discounts the importance of ideology and culture and sees man as essentially a rational, profit-maximizing individual. But one small example will illustrate the problematic character of such materialist views.

Max Weber begins his famous book, *The Protestant Ethic and the Spirit of Capitalism*, by noting the different economic performance of Protestant and Catholic communities throughout Europe and America, summed up in the proverb that Protestants eat well while Catholics sleep well. Weber noted that in many traditional peasant communities, raising the piece-work rate had the opposite effect of *lowering* labor productivity: at the higher rate, a peasant accustomed to earning two and one-half marks per day found he could earn the same amount by working less, and did so because he valued leisure more than income. The choices of leisure over income, or of the militaristic life of the Spartan hoplite over the wealth of the Athenian trader, [...] cannot possibly be explained by the impersonal working of material forces, but come preeminently out of the sphere of consciousness. A central theme of Weber's work was to prove that contrary to Marx, the material mode of production, far from being the "base," was itself a "superstructure" with roots in religion and culture, and that to understand the emergence of modern capitalism and the profit motive one had to study their antecedents in the realm of the spirit.

Q1. According to the Hegelian view, the roots of all ideology lie in

- ☐ a) religious doctrines.
- ☐ b) cultural norms.
- ☐ c) moral habits.
- ☒ d) consciousness. ✓ Your answer is correct

Q2. It can be inferred from the passage that people on the Right would

- ☒ a) **subscribe to materialism, like the people on the Left.** ✓ **Your answer is correct**
- ☐ b) advocate the importance of consciousness, unlike the people on the Left.
- ☐ c) support the materialist bias of modern thought, as opposed to the people on the Left.
- ☐ d) **endorse the importance of ideology and culture, just like the people on the Left.**

Q3. What is the primary concern of the author in the passage?

- ☐ a) **To dismiss the doctrine of idealism under the realm of materialism as the prevailing material mode of production in a given society controls its collective consciousness.**
- ☐ b) To stress the deterministic role played by consciousness in shaping society.
- ☐ c) To demonstrate the complexity of relating the philosophy of the ideal world to that of the real world.
- ☐ d) **To prove that societal consciousness assumes the form of religious, cultural and moral attributes.**

Q4. All of the following can be understood from the passage EXCEPT?

- ☐ a) **At least one materialist theory posits man as a rational, profit-maximising individual.**
- ☐ b) Hegel believed that one may not be entirely aware of how consciousness manifested in one's life.
- ☐ c) Paul Kennedy believed that great empires declined because of economic reasons.
- ☐ d) **Hegel believed that the material world could not infringe on the conscious world that prevailed over it.**

DIRECTIONS *for questions 5 to 8:* The passage given below is accompanied by a set of **four** questions. Choose the best answer to each question.

The American evolutionary theorist Lynn Margulis, who popularised the theory of symbiosis, first coined the term 'holobiont' in 1991. She was interested in long-term, tightly integrated associations such as those evident in lichens – the crusty-looking growths found on rocks and trees, made of fungus conjoined with algae. Margulis thought that there was a tight analogy between an egg and a sperm coming together to form a new organism, and the coming together of two species to form a new symbiotic consortium, which she called a holobiont.

Margulis argued that the interactions within a holobiont aren't too different from the life cycle of sexually reproducing organisms. The partners are integrated wholes that die and reproduce as one. But instead of sending out tiny cells to reproduce, holobionts send out individual organisms of different species.

With this framing in mind, when biologists began to use the term in the 1990s, they applied it to a few (usually two) organisms. But the word took on a very different cast in the hands of the American coral reef biologist Forest Rohwer and his colleagues, who defined a holobiont as a host and all of its associated microorganisms.

Two protagonists just aren't enough when it comes to explaining the evolutionary success of corals. They are made up of clusters of polyps, tiny wiggling things that get by with just a few tentacles and a toothless maw. Coral polyps reproduce by cloning themselves, and then sticking together to form large colonies, supported by a jointly fashioned skeleton. The most spectacular corals work hand-in-hand with photosynthetic algae that they host within their own cells. The algae provide nutrients via photosynthesis, while the coral gives the algae both food and protection. And those simple little polyps don't end their symbiotic relationships there. Corals don't possess a complicated immune system to fend off pathogens; instead, they seem to selectively cultivate helpful or benign bacteria, which crowd out the harmful microbes. Corals also produce mucus that appears to be able to trap phages, viruses that infect and kill only bacteria. An enemy of an enemy is a friend, after all.

Rohwer and colleagues, unaware of Margulis's idea, introduced the term holobiont to capture the dynamics of coral physiology. As a result, by the early 2000s, the scientific literature contained two contrasting definitions. One picked out an organism-like symbiotic pair that reproduced, while the other identified an ecological community of microbes indexed to a host.

For a time, the ecological account prevailed. But Margulis's physiological conception of holobionts was revitalised in the late 2000s as part of a new theory: what's known as the *hologenome theory of evolution*. Advocates merged *both* versions of holobiont into something a bit more conceptually loaded. On this view, the ecological notion of holobiont (the host and all its resident microbes) is given additional properties. It's an entity that's coherent enough to have its own hologenome, made up of the host genome plus all the microbial genomes. A major implication of this theory is that natural selection doesn't just act on the genome of individual organisms: it acts on the hologenome of holobionts, which are seen as single units that can evolve at the level of the holobiont.

Q5. How is the holobiont as defined by Rohwer different from that defined by Margulis?

☒ a) Margulis's holobiont is a symbiotic union between a pair of different species hosting another organism while Rohwer's holobiont is a community of microorganisms that reproduce by sending out tiny cells. **✖**

Your answer is incorrect

☐ b) Margulis's holobiont involves a symbiotic association between two different species, one hosted by another, while Rohwer's holobiont is a system where there is one host and more than one other organism coexisting together.

☐ c) Margulis's holobiont is the symbiotic consortium formed when two different species of organisms die and reproduce as a whole, while Rohwer's holobiont is formed when different organisms of the same species reproduce by cloning themselves.

☐ d) Margulis's holobiont is a symbiotic union between organisms belonging to two different species, while Rohwer's holobiont is a system formed by a host and multiple other organisms associated with it.

Q6. The author mentions the phrase, “an enemy of an enemy is a friend, after all”, in order to

- ☐ a) **establish that the symbiotic relationship between the corals and benign bacteria acts as a defense mechanism of the holobiont.**
- ☐ b) illustrate the benefits that corals gain from helpful bacteria.
- ☐ c) highlight the advantage that bacteria derive from the relationship with corals.
- ☐ d) **illustrate the advantage that corals provide to virus that can help kill bacteria that are harmful to the corals.**

Q7. All of the following statements about corals are definitely true EXCEPT:

- ☐ a) **The definition of holobiont by Margulis does not comprehensively explain the evolutionary success of corals.**
- ☐ b) Most corals have photosynthetic algae, which receive food and protection from the former.
- ☐ c) The mode of reproduction that coral polyps resort to is cloning themselves.
- ☐ d) **The immune system of the corals is not enough to ward off harmful microbes that threaten the corals.**

Q8. Which of the following is a takeaway from the hologenome theory of evolution?

- ☐ a) **Evolutionary forces act on a host genome and all the microbial genomes of a holobiont.**
- ☐ b) Natural selection does not act on the genome of individual microorganisms in a holobiont.
- ☐ c) Natural selection acting on a holobiont depends on how similar the host genome is to the hologenome.
- ☐ d) **Natural selection plays a more important role in the evolution of a holobiont than that of an individual organism.**

DIRECTIONS *for questions 9 to 12:* The passage given below is accompanied by a set of **four** questions. Choose the best answer to each question.

Timothy Fridtjof "Tim" Flannery (born 28 January 1956) is an Australian mammalogist, palaeontologist, environmentalist and global warming activist. ... His 2005 book, "The Weather Makers", about climate science and global warming, was a bestseller. His latest is his most ambitious book so far. The publisher calls "Here on Earth" a twin biography, of humanity and the planet it inhabits, but that description is inadequate. Mr Flannery's subject is the likely fate of humankind, and whether the powers granted to modern civilisation by science and technology will prove to be its downfall or its salvation.

He muses on whether humanity counts as a superorganism (a classification usually reserved for bees and ants), why we have yet to discover intelligent aliens, the poorly understood effects of dumping industrial chemicals into the environment, the power of planet-watching networks of satellites and the benefits of aboriginal scrub-burning. There is an effort to organise the chapters around two competing models of human behaviour, a co-operative, far-sighted wisdom that Mr Flannery dubs the "Gaian" approach and a reckless, ultimately destructive short-termism that he calls "Medean", after the bloodthirsty enchantress of Greek myth.

The trouble is that the subject is far too big to fit comfortably into a book a little more than 300 pages long. Mr Flannery is a respected biologist with plenty of published papers to his name, but the book feels dilettantish, with a dizzying array of concepts introduced, briefly discussed, then dispensed with before the reader has had time to digest them. The question of whether modern democracies can successfully resist populist tyranny is raised and then abandoned within two paragraphs. A discussion of decarbonising the world's transport networks flashes by in a handful of pages. Fewer examples more deeply explored might have added up to a more convincing case, and an easier read.

That is a shame, because much of the material would repay a deeper look. The "Gaia" theory, which is the subject of the opening chapters, starts with the surprising observation that such things as the Earth's surface temperature and the salinity of its oceans have fluctuated remarkably little over billions of years despite, for instance, a significant increase in the power of the sun. James Lovelock, who first proposed the Gaia theory, used an analogy with living organisms, all of which can, to some degree, regulate conditions within their own bodies. The theory brings to mind New Age crankery, and some of the research Mr Flannery cites to underpin it is speculative at best. For example, he mentions a scientific paper that suggests that lichens, bacteria and the like may have been a driving force behind prehistoric geology and the formation of continents. Mr Flannery admits that the idea is "controversial". But having shown the reader a glimpse of this fascinating byway, he speeds straight past, impatient to reach the next intellectual stop, while the tentative theory is simply accepted.

If the book is not all it could be, it is still worth reading, though less for answers than for its interesting hypotheses.

Q9. Which of the following is inconsistent with the information presented in the passage?

- ☐ a) **The Gaia Theory essentially deals with whether the biosphere can regulate the environment to suit itself.**
- ☐ b) Mr. Flannery succeeds in thumbing down the fundamentals of the "Gaia theory" in his book.
- ☐ c) The "Gaia Theory" forms an introductory part of the book "Here on Earth".
- ☐ d) **The Gaian approach adopted by Flannery helps draw attention to positive, cooperative possibilities in human behaviour.**

Q10. Which of the following about the book “Here an Earth” is not true?

- a. The book focuses on what man might do to himself and his environment through the power he has acquired.
- b. This is the first book of the author to become popular and make him well-known.
- c. The book provides a few hypotheses without explaining them in totality.
- d. The book looks at man in two contradictory roles.
- e. It portrays the author as being a little wary because of the confusion and contradictions in the topic of discussion and unwilling to take sides in a debatable issue.

- ☐ a) **b and e**
- ☐ b) c, d and e
- ☐ c) a and e
- ☐ d) **b and d**

Q11. Tim Flannery, as per the passage, is all of the following EXCEPT?

- a. A person who studies living beings, life processes, mammals and past geological periods as revealed by fossil remains.
- b. An author who comes across as 'rambling' to a reader perusing his book "Here on Earth".
- c. A person who wonders whether the planet can be saved through science and technology.
- d. A person who, though unsure about the validity of the Gaian and Medean approaches, uses them anyway, to explain the larger picture on Earth.

- ☐ a) **Only d**
- ☐ b) a and b
- ☐ c) c and d
- ☐ d) **b and d**

Q12. What, according to the reviewer, are the shortcomings of the book "Here on Earth"?

- ☐ a) **The book deals perfunctorily with a wide range of topics.**
- ☐ b) The book presents two models of humans that are at odds with each other.
- ☐ c) The book fails to account for man's inability to discover aliens.
- ☐ d) **The book is difficult to read as it has only a few examples to illustrate difficult concepts.**

DIRECTIONS *for questions 13 to 16:* The passage given below is accompanied by a set of **four** questions. Choose the best answer to each question.

Architecture is about evolution, not revolution. It used to be thought that once the Romans pulled out of Britain in the fifth century, their elegant villas and carefully-planned towns simply fell into decay as British culture was plunged into the Dark Ages. It took the Norman Conquest of 1066 and the Gothic cathedral-builders of the Middle Ages to bring back the light and revive the British culture.

However, the truth is not as simple as that. Romano-British culture, including architecture, survived long after the Roman withdrawal. And although the Anglo-Saxons had a sophisticated building style of their own, little survives to bear witness to their achievements as the vast majority of Anglo-Saxon buildings were made of wood.

The period between the Norman landing at Pevensey in 1066 and the day in 1485 marks a rare flowering of British building. ... The underlying ethos of medieval architecture was 'fitness for purpose'. The great cathedrals were fiercely functional buildings. Castles served their purpose and their battlements were for use rather than ornament.

The buildings of the 16th century were also governed by fitness for purpose but the purpose was very different. In domestic architecture, buildings were used to display status and wealth. A greater sense of security led to more outward-looking buildings, as opposed to the medieval arrangement where the need for defence created houses that faced inward onto a courtyard or series of courtyards. This allowed for much more in the way of exterior ornament. ... There was also a general move towards balanced and symmetrical exteriors with central entrances. In addition, there was progress towards more stable and sophisticated houses for those lower down the social scale. Stone, and later brick, began to replace timber as the standard building material for the homes of farmers and artisans.

In spite of this building boom, the Renaissance was generally slow to arrive in England, largely because Elizabeth's troublesome relations with Catholic Europe made the free exchange of ideas difficult. Craftsmen and pattern-books did come over from the Protestant Low Countries, but our relative isolation from the European cultural mainstream led to a national style which was a bizarre though attractive mixture of Gothic and classical styles.

But during the 1640s and 50s, the Civil War sent many people to the Continent either to escape the fighting or, when the war was lost, to follow Charles II into exile. There they came into contact with French, Dutch and Italian architecture and, with Charles's restoration in 1660, there was a flurry of building activity as royalists reclaimed their property and built themselves houses reflecting the latest European trends.

As the century wore on, this resolved itself into a passion for the Baroque grandeur which Louis XIV had turned into an instrument of statecraft at Versailles. Formal, geometrical and symmetrical planning meant that a great lord could sit in his dining chamber, at the physical as well as the metaphorical centre of his world, with suites of rooms radiating out in straight lines to either side. ...

The British Baroque was a reassertion of authority, an expression of absolutist ideology by men who remembered a world turned upside down during the Civil War. The style is heavy and rich, sometimes overblown and melodramatic.

Q13. According to the passage, why was the Renaissance slow to arrive in England?

- ☐ a) **There was a dearth of architects, craftsmen and pattern-books.**
- ☒ b) **Queen Elizabeth had a worrisome association with Catholic Europe impeding the free interchange of ideas.** ✓ **Your answer is correct**
- ☐ c) Rules of architecture in Elizabeth's reign were archaic and stringent and they cramped the genius of the architect masters.
- ☐ d) **The 'fitness for purpose' ethos of medieval architecture continued to reign supreme in the "building boom" of the 16th century.**

Q14. The second para of the passage serves which of the following purposes?

- ☒ a) It presents an alternative hypothesis and provides arguments that support it. **✗ Your answer is incorrect**
- ☐ b) It challenges the validity of a theory by exposing the inconsistencies and contradictions in it.
- ☐ c) It explains an implicit idea by reiterating a point.
- ☐ d) It arrests the reader's attention and clears a misconception by providing an alternative explanation.

Q15. Which of the following is true about the architectural style of the 16th century?

- ☐ a) In the 16th century, a greater sense of security led to more outward-looking buildings heightening the exterior embellishment on display.
- ☒ b) In the 16th century, English architecture deflected from the objective of 'fitness for purpose'. **✗ Your answer is incorrect**
- ☐ c) A dreary mixture of gothic and classical styles characterized English architecture in the 16th century.
- ☐ d) In the 16th century, royalists borrowed architectural ideas from the French, the Dutch and the Italians.

Q16. With reference to the Baroque style as discussed in the passage, which of the following statements is incorrect?

- ☐ a) **The Palace of Versailles was executed in the Baroque style.**
- ☐ b) **The British Baroque style was a rebellion against the cataclysm of the Civil War.**
- ☐ c) The Baroque style involved formal, geometrical and symmetrical planning and the style itself was rich, weighty and sensational.
- ☐ d) **The Baroque style was a union of a number of architectural styles.**

Q17. DIRECTIONS *for question 17:* The sentences given in the question, when properly sequenced, form a coherent paragraph. Each sentence is labelled with a number. Decide on the proper order for the four sentences and key in the sequence of four numbers as your answer, in the input box given below the question.

1. The idea that only work that pushes boundaries is really worth reading seems troubling.
2. Perhaps the real radicalism in poetry lies in acknowledging and valuing more conservative forms alongside more current examples of resistance, if we are really interested in diversity, that is.
3. More sensible, I think, to recognise these poems simply as different, as different as, say, Alice Oswald's expansive Dart from her first collection of largely single-page poems, The Thing in the Gap Stone Stile.
4. In poems such as Spelt from Sybil's Leaves, Gerard Manley Hopkins wrestled with and smashed open the sonnet form but, as wonderful and strange as those early sonnets are, can one definitively say they are better, more powerful than The Terrible Sonnets of his final years, in which the conventions of the form reasserted themselves?

Q18. DIRECTIONS *for question 18:* The passage given below is followed by four alternate summaries. Choose the option that best captures the essence of the passage.

Nutrient composition and caloric intake have traditionally been used to devise optimized diets for various phases of life. Adjustment of meal size and frequency have emerged as powerful tools to ameliorate and postpone the onset of disease and delay aging, whereas periods of fasting, with or without reduced energy intake, can have profound health benefits. Future research endeavours should be directed to the integration of a balanced nutritious diet with controlled meal size and patterns and periods of fasting to develop better strategies to prevent, postpone, and treat the socioeconomical burden of chronic diseases associated with aging.

- ☐ a) **The cost of age-related diseases can be effectively tackled by combining the benefits of fasting with those of a balanced diet.**
- ☐ b) **It is worthwhile to invest in developing an effective regimen combining balanced diet, fasting, and controlled meal size to ameliorate the negative impact of geriatric diseases.**
- ☐ c) Research on exposing people suffering from age-related diseases to the benefits of combining a good diet with intermittent fasting is lacking and is the need of the hour.
- ☐ d) **Older people can postpone, or even avoid, the onset of diseases through diet control alternating with periods of fasting.**

Q19. DIRECTIONS *for question 19:* The sentences given in the question, when properly sequenced, form a coherent paragraph. Each sentence is labelled with a number. Decide on the proper order for the four sentences and key in the sequence of four numbers as your answer, in the input box given below the question.

1. But bright light from billboards, streetlights and houses is interfering and blocking potential firefly couples from pairing up.

2. It's getting harder for them to reproduce because light pollution is outshining their mating signals.

3. Males light up to signal availability and females respond with patterned flashes to show that they're in the mood.

4. Fireflies, it turns out, use their special glowing powers in courtship.

Q20. DIRECTIONS *for question 20:* There is a sentence that is missing in the paragraph below. Look at the paragraph and decide in which blank (option 1, 2, 3, or 4) the following sentence would best fit.

Sentence: March of 1998 marked the start of a cult following and a lifestyle change for many people.

Paragraph: It was not the experts in cinema that kept this film alive, but the avid movie-goers who saw something more than a lazy and lucky main character who lived in a constant haze. ____ (1) ____ The Big Lebowski has lived on thanks to these followers. It has become a cult movie, like Animal House and The Goonies before it, revamping the love for a certain creamy cocktail and inspiring a whole new quasi-religion: Dudeism. ____ (2) ____ The followers of Dudeism practice Lebowski's philosophy on life: to go with the flow and stay calm, no matter what hits you. ____ (3) ____ It is easy to list the ways Lebowski has influenced today's culture, with White Russian competitions and festivals that sport statues of The Dude. ____ (4) ____ But going further than just the tangible evidence, this Coen brothers' masterpiece has given us a definitive make-or-break in relationships. If they don't like The Big Lebowski, it's safe to say they're not a keeper.

- ☐ a) **Option 1**
- ☐ b) **Option 2**
- ☐ c) Option 3
- ☐ d) **Option 4**

Q21. DIRECTIONS *for questions 21 and 22:* The passage given below is followed by four alternate summaries. Choose the option that best captures the essence of the passage.

Traditionally, a "belief" is a state of mind of a certain sort. But the behaviourists deny that there are states of mind, or at least that they can be known; they therefore avoid the word "belief", and, if they used it, would mean by it a characteristic of bodily behaviour. There are cases in which this usage would be quite in accordance with common sense. Suppose you set out to visit a friend whom you have often visited before, but on arriving at your destination you find that he has moved, you would say "I thought he was still living at his old house." Yet it is highly probable that you did not think about it at all, but merely pursued the usual route from habit. A "thought" or "belief" may, therefore, in the view of common sense, be shown by behaviour, without any corresponding "mental" occurrence. And even if you use a form of words such as is supposed to express belief, you are still engaged in bodily behaviour, provided you pronounce the words out loud or to yourself. Shall we say, in such cases, that you have a belief? Or is something further required?

- ☐ a) **There is nothing called mind – it is all a complex interplay of neurons which are physical in nature.**
- ☐ b) **Behaviourists consider belief to be a behavioural characteristic of body rather than mind.**
- ☐ c) Behaviour acts at two different levels – body and mind and both are significant in contributing to a man's actions.
- ☐ d) **Belief does not require any mental thought behind it and can manifest itself by actions.**

Q22. DIRECTIONS *for questions 21 and 22:* The passage given below is followed by four alternate summaries. Choose the option that best captures the essence of the passage.

A small world network is a network structure where there are lots of clusters but there are also connections or bridges between these clusters. You can imagine a small world network to be somewhere in between random and clustered network. It's clustered but with random connections between clusters. In pre-Internet days, you'd spent most of your time hanging out with the same colleagues and friends, but whenever you went to a party or a conference, you bumped into people that expanded your worldview. In this way, the conference or party became your bridge to other clusters. In the tech world, email is an example of a small world network (you mostly exchange emails with known people but every now and then have an interaction with a stranger). Another example would be Twitter where you are connected with both people you know and those you don't know.

- ☐ a) **There are several examples of small world networks around us from Twitter to email.**
- ☐ b) **A small world network encompasses a group of randomly connected clusters.**
- ☐ c) Small world networks are neither random, nor organised, and are midway between the two.
- ☐ d) **A combination of clustered networks in a real world gives us a small world network.**

Q23. DIRECTIONS *for question 23:* There is a sentence that is missing in the paragraph below. Look at the paragraph and decide in which blank (option 1, 2, 3, or 4) the following sentence would best fit.

Graphene got its beginnings in 2003, when scientists at the University of Manchester found they could peel off a gossamer film of the material just by touching a piece of ordinary sticky tape to a block of purified graphite.

The wonder material graphene — an array of interlinked carbon atoms arranged in a sheet just one atom thick — promised a world of applications, including super-fast electronics, ultra-sensitive sensors and incredibly durable materials. _____(1)_____ After a few false starts, that promise is close to realization. _____(2)_____ And a suite of other extremely thin substances is following in its wake. _____(3)_____ Graphene proved stronger than steel but extremely flexible, and electrons could zip through it at high speeds. _____(4)_____ It earned its discoverers the Nobel Prize in 2010, but researchers spent years struggling to manufacture it on larger scales and figuring out how its remarkable properties could best be used.

☐ a) **Option 1**

☐ b) **Option 2**

☐ c) **Option 3**

☐ d) **Option 4**

Q24. DIRECTIONS *for question 24:* The sentences given in the question, when properly sequenced, form a coherent paragraph. Each sentence is labelled with a number. Decide on the proper order for the four sentences and key in the sequence of four numbers as your answer, in the input box given below the question.

1. With his small team, he travelled to the remote island of Príncipe off the western coast of Africa to observe the longest total solar eclipse — 6 minutes and 51 seconds — in five centuries.
2. When the Moon curtained the sun, Eddington hoped to see light of the Hyades cluster positioned directly behind the sun from Earth's vantage point.
3. At a time when very few scientists considered relativity plausible, and very few Englishmen would risk their reputation by defending a German's ideas, Eddington set out to test Einstein's theory against reality in an ingenious experiment nature herself had furnished.
4. If Einstein was right and Newton wrong, the sun's massive gravitational field would warp spacetime itself, bending the path of the light to make it visible from Earth, and the Hyades starlight would, thus, be deflected from its baseline night-time position.